

CHEMISTRY**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 8 printed pages.
4. It is divided into four sections and has twenty one questions in all.
5. Answer all questions.
6. Section A has fourteen subparts. Each question carries 1 mark.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B has ten questions. Each question carries 2 marks.
9. Section C has seven questions. Each question carries 3 marks.
10. Section D has three questions. Each question carries 5 marks.
11. Internal choices have been provided in two questions each in Sections B, C and D.
12. The intended marks for questions are given in brackets [].
13. All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.
14. Balanced equations must be given wherever possible and diagrams where they are helpful.
15. When solving numerical problems, all essential workings must be shown.
16. In working out problems, use the following data:
 Gas constant $R = 1.987 \text{ cal deg}^{-1} \text{ mol}^{-1} = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$, $1 \text{ l atm} = 1 \text{ dm}^3 \text{ atm} = 101.3 \text{ J}$.
 1 faraday = 96500 coulombs, Avogadro's number = 6.023×10^{23} .

Instruction to Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

SECTION A – 14 MARKS**Question 1**

- (A) Fill in the blanks by choosing the appropriate word(s) from those given in the brackets.
 [Clemmensen, Wolff-Kishner, Rosenmund, Kohlrusch's, Faraday's, increases, decreases, lanthanoid contraction, actinoid contraction, ethoxyethane, methoxyethane, Williamson, Kolbe's, Etard, anode, cathode]
- (i) Acetophenone is reduced to ethylbenzene by zinc amalgam and concentrated HCl in _____ reduction, while the same change brought about by hydrazine and KOH in ethylene glycol is called _____ reduction. (Recall) [1]

- (ii) The limiting molar conductivity of a weak electrolyte cannot be found by extrapolation; it is obtained by applying _____ law. The molar conductivity of every electrolyte _____ on dilution. (*Understand*) [1]
- (iii) The steady decrease in atomic and ionic size from La to Lu is called _____; as a result the basic character of the hydroxides from $\text{La}(\text{OH})_3$ to $\text{Lu}(\text{OH})_3$ _____. (*Understand*) [1]
- (iv) Sodium ethoxide reacts with bromoethane to form _____; the reaction is known as _____ synthesis. (*Recall*) [1]
- (B) Select and write the correct alternative from the choices given below.
- (i) Consider the following statements about aqueous solutions at the same temperature: [1]
 P: 0.1 m NaCl has a higher boiling point than 0.1 m glucose.
 Q: 0.1 m K_2SO_4 has a lower freezing point than 0.1 m NaCl.
 R: 0.1 M urea and 0.1 M glucose have the same osmotic pressure.
 S: The relative lowering of vapour pressure of 0.1 m sucrose is greater than that of 0.1 m KCl.
 The correct statements are:
 (*Evaluate*)
- (a) P and Q only
 (b) P, Q and R only
 (c) Q, R and S only
 (d) P and S only
- (ii) The base present in RNA but not in DNA is: (*Recall*) [1]
 (a) thymine
 (b) uracil
 (c) adenine
 (d) cytosine
- (iii) Which of the following complexes is tetrahedral and paramagnetic? (Atomic number of Ni = 28, Zn = 30) (*Analyse*) [1]
 (a) $[\text{Ni}(\text{CO})_4]$
 (b) $[\text{NiCl}_4]^{2-}$
 (c) $[\text{Ni}(\text{CN})_4]^{2-}$
 (d) $[\text{Zn}(\text{NH}_3)_4]^{2+}$
- (iv) Which of the following alkyl bromides undergoes $\text{S}_{\text{N}}1$ hydrolysis most rapidly? (*Analyse*) [1]
 (a) CH_3Br
 (b) $\text{CH}_3\text{CH}_2\text{Br}$
 (c) $(\text{CH}_3)_2\text{CHBr}$
 (d) $(\text{CH}_3)_3\text{CBr}$
- (v) The correct order of basic strength in aqueous solution is: (*Evaluate*) [1]
 (a) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > \text{C}_6\text{H}_5\text{NH}_2$
 (b) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{C}_6\text{H}_5\text{NH}_2$
 (c) $\text{C}_6\text{H}_5\text{NH}_2 > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$
 (d) $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{C}_6\text{H}_5\text{NH}_2 > (\text{CH}_3)_2\text{NH}$

- (vi) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: Ethanol has a lower boiling point than its isomer methoxymethane.

Reason: Methoxymethane molecules are associated through intermolecular hydrogen bonding.

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.
- (vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: In the electrolysis of aqueous NaCl, hydrogen and not sodium is liberated at the cathode.

Reason: The reduction potential of water (to H_2) is much higher than that of Na^+/Na .

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.
- (C) Read the passage carefully and answer the questions that follow.

Many reactions, including every radioactive decay, follow first-order kinetics: the rate is proportional to the amount of reactant present, and the half-life, $t_{1/2} = 0.693/k$, does not depend on the initial amount. Carbon-14 ($t_{1/2} = 5730$ years) is made continuously in the upper atmosphere and is taken up by living plants; once the plant dies, no fresh carbon-14 enters and the activity of its remains falls steadily. Comparing the activity of an old sample with that of fresh wood gives its age.

The rate constant of a reaction also depends on temperature, as described by the Arrhenius equation, $k = A e^{-E_a/RT}$. A graph of $\log k$ against $1/T$ is a straight line from which E_a can be found.

- (i) How many half-lives are needed for 87.5% of a radioactive sample to decay? (*Apply*) [1]
(ii) A wooden object from an archaeological site shows one-fourth of the carbon-14 activity of fresh wood. Estimate its age. (*Apply*) [1]
(iii) What is the slope of the graph of $\log k$ against $1/T$? (*Understand*) [1]

SECTION B – 20 MARKS

Question 2

On diluting from 0.1 M to 0.001 M, the molar conductivity of KCl increases about 1.1 times, whereas that of acetic acid increases about 9 times. Explain this difference and show it on a graph of Λ_m against $\sqrt{c}$. (*Analyse*) [2]

Question 3

An aromatic compound [A], C_7H_6O , gives a silver mirror with Tollens' reagent but does not reduce Fehling's solution. On heating with concentrated NaOH, [A] gives [B] and [C]. [C] is a sodium salt which, on heating with soda lime, gives benzene. Identify [A], [B] and [C] and write the equation for the reaction of [A] with concentrated NaOH. (*Understand*) [2]

Question 4

- (i) Answer the following with respect to the complex ion $[Ni(CN)_4]^{2-}$. (Atomic number of Ni = 28)
- (a) What is the hybridisation of the central metal ion and the geometry of the complex? (*Understand*) [1]
- (b) Comment on its magnetic nature. (*Analyse*) [1]

OR

- (ii)
- (a) Draw the crystal field splitting of d orbitals in a tetrahedral field. (*Understand*) [1]
- (b) Why are low-spin tetrahedral complexes rarely observed? (*Analyse*) [1]

Question 5

A first-order reaction is 40% complete in 25 minutes. Calculate the time required for 80% completion. (*Apply*) [2]

Question 6

How can the following conversions be carried out? (Write the chemical equations.)

- (i) Aniline to phenol (*Recall*) [1]
- (ii) Phenol to benzene (*Recall*) [1]

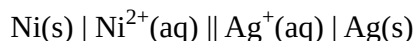
Question 7

Substantiate the following statements with reasons.

- (i) Cu^+ compounds are colourless and diamagnetic, whereas Cu^{2+} compounds are coloured and paramagnetic. (Atomic number of Cu = 29) (*Understand*) [1]
- (ii) Transition metals form a large number of complex compounds. (*Understand*) [1]

Question 8

Calculate the standard emf and the standard Gibbs energy change for the cell [2]



and state whether the cell reaction is spontaneous.

Given: $E^\circ(Ni^{2+}/Ni) = -0.25\text{ V}$, $E^\circ(Ag^+/Ag) = +0.80\text{ V}$ (*Apply*)

Question 9

- (i) Illustrate the following reactions with an example in each case:
- (a) Carbylamine reaction (*Recall*) [1]

- (b) Sandmeyer reaction (*Recall*) [1]

OR

- (ii) An amine [A], C_2H_7N , reacts with benzenesulphonyl chloride to give [B], which is insoluble in alkali. An isomer [C] of [A] gives a foul-smelling compound with chloroform and alcoholic KOH, and reacts with benzenesulphonyl chloride to give [D], which is soluble in alkali. Write the structures of [A], [B], [C] and [D]. (*Understand*) [2]

Question 10

Aman and Tina were given two unlabelled samples, methanol and ethanol. Aman proposed to tell them apart by the Lucas test; Tina proposed the iodoform test.

- (i) Which proposal is reliable? Give reasons with observations. (*Evaluate*) [1]
 (ii) Write the equation for the positive test in Tina's case. (*Apply*) [1]

Question 11

In a hill town the car radiators are filled with a solution of ethylene glycol ($C_2H_6O_2$) in water. [2]
 What mass of ethylene glycol must be added to 5 kg of water to lower its freezing point to $-2.8^\circ C$? (K_f of water = $1.86 K kg mol^{-1}$, molar mass of ethylene glycol = $62 g mol^{-1}$) (*Apply*)

SECTION C – 21 MARKS

Question 12

Three students were given 1-bromopropane. Meera heated it with aqueous KOH, Kabir with alcoholic KOH and Sana with alcoholic silver cyanide. Write the chemical equation of each reaction and the IUPAC name of the organic product obtained by:

- (i) Meera (*Understand*) [1]
 (ii) Kabir (*Understand*) [1]
 (iii) Sana (*Understand*) [1]

Question 13

The following data were obtained for the reaction $2NO + Br_2 \rightarrow 2NOBr$:

Run	[NO] ($mol L^{-1}$)	[Br ₂] ($mol L^{-1}$)	Initial rate ($mol L^{-1} s^{-1}$)
1	0.10	0.10	1.2×10^{-4}
2	0.20	0.10	4.8×10^{-4}
3	0.20	0.30	1.44×10^{-3}

- (i) Deduce the rate law. (*Analyse*) [1]
 (ii) Calculate the rate constant with its unit. (*Apply*) [1]
 (iii) Calculate the initial rate when $[NO] = 0.15 mol L^{-1}$ and $[Br_2] = 0.25 mol L^{-1}$. (*Apply*) [1]

Question 14

- (i) Arrange ethanol, phenol, water and p-nitrophenol in increasing order of acidic strength. Give reasons for the position of ethanol and of p-nitrophenol. (*Evaluate*) [2]

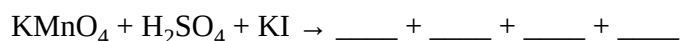
- (ii) Why should ethers not be distilled to dryness? (*Understand*) [1]

OR

- (iii) How will the following be converted? (Give chemical equations.)
- (a) Phenol to salicylic acid (*Recall*) [1]
- (b) Ethanal to propan-2-ol (*Recall*) [1]
- (c) Ethanol to ethoxyethane (*Recall*) [1]

Question 15

- (i) Complete and balance the following chemical equation: [1]



(*Understand*)

- (ii) What is observed when an alkali is added to an aqueous solution of potassium dichromate? Write the ionic equation. (*Apply*) [1]
- (iii) Cr^{2+} is a reducing agent whereas Mn^{3+} is an oxidising agent, though both are d^4 . Explain. (Atomic number of Cr = 24, Mn = 25) (*Analyse*) [1]

Question 16

- (i) What is formed when glucose is treated with bromine water? What does this reaction indicate about the structure of glucose? (*Understand*) [1]
- (ii) Write the structure of the dipeptide glycylalanine and mark the peptide bond. (*Understand*) [1]
- (iii) State one structural difference between DNA and RNA. (*Recall*) [1]

Question 17

- (i) Aluminium is obtained by the electrolysis of alumina dissolved in molten cryolite. (Atomic mass of Al = 27)
- (a) How many coulombs of electricity are needed to produce 2.70 g of aluminium? (*Apply*) [2]
- (b) How long would this take with a current of 5.0 A? (*Apply*) [1]

OR

- (ii) The conductivity of a 0.00241 M acetic acid solution is $7.896 \times 10^{-5} \text{ S cm}^{-1}$ at 298 K.
- (a) Calculate its molar conductivity. (*Apply*) [2]
- (b) If Λ°_{m} for acetic acid is $390.5 \text{ S cm}^2 \text{ mol}^{-1}$, what is its degree of dissociation? (*Apply*) [1]

Question 18

Write the chemical equations for the following conversions: (*Recall*)

- (i) Nitrobenzene to aniline [1]
- (ii) Ethylamine to ethanol [1]
- (iii) Aniline to 2,4,6-tribromoaniline [1]

SECTION D – 15 MARKS

Question 19

- (i) Answer the following:
- (a) A solution containing 3.00 g of a protein in 500 mL of water has an osmotic pressure of 1.64×10^{-3} atm at 27 °C. Calculate the molar mass of the protein. (*Apply*) [2]
- (b) A solution of 12.48 g of barium chloride in 1000 g of water boils at 373.0832 K. Calculate the degree of dissociation of barium chloride. (K_b of water = $0.52 \text{ K kg mol}^{-1}$, molar mass of $\text{BaCl}_2 = 208 \text{ g mol}^{-1}$, boiling point of water = 373 K) (*Apply*) [2]
- (c) Why are soda-water bottles sealed under high pressure? (*Understand*) [1]

OR

- (ii)
- (a) The vapour pressure of water at 298 K is 23.8 mm Hg. Calculate the vapour pressure of a solution of 5.0 g of urea (molar mass 60 g mol^{-1}) in 100 g of water at this temperature. (*Apply*) [2]
- (b) A 5% (w/v) solution of cane sugar (molar mass 342 g mol^{-1}) is isotonic with a 0.877% (w/v) solution of a non-electrolyte X at the same temperature. Calculate the molar mass of X. (*Apply*) [2]
- (c) What happens when red blood cells are placed in 0.5% (w/v) NaCl solution? (*Understand*) [1]

Question 20

- (i) An organic compound [A], $\text{C}_2\text{H}_4\text{O}$, reduces Tollens' reagent and gives the iodoform test. [A] reacts with HCN and the product on hydrolysis gives [B], $\text{C}_3\text{H}_6\text{O}_3$. [A] with dilute NaOH gives [C], which on heating gives [D], $\text{C}_4\text{H}_6\text{O}$. Identify [A], [B], [C] and [D]. (*Understand*) [2]
- (ii) Write the equation for the reaction of ethanal with hydroxylamine. Name the product. (*Recall*) [1]
- (iii) Arrange CH_3COOH , ClCH_2COOH , Cl_2CHCOOH and FCH_2COOH in increasing order of acid strength. (*Evaluate*) [1]
- (iv) Give a chemical test to distinguish benzaldehyde from ethanal. (*Analyse*) [1]

Question 21

- (i) Answer the following questions with respect to the complex ion $[\text{Co}(\text{en})_2\text{Cl}_2]^+$.
- (a) Write its IUPAC name. (*Recall*) [1]
- (b) Draw its geometrical isomers and state which of them is optically active. (*Understand*) [2]
- (c) Write the t_{2g}/e_g configuration of the central ion and comment on the magnetic nature of the complex. (Atomic number of Co = 27) (*Analyse*) [1]
- (d) Name the type of isomerism shown by $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}_2$. (*Analyse*) [1]

OR

- (ii)
- (a) Three isomers of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ are violet, light green and dark green. One mole of each precipitates 3, 2 and 1 mole of AgCl respectively with excess AgNO_3 . Write their formulae and name the type of isomerism. (*Evaluate*) [2]
- (b) Compare $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ with respect to type of hybridisation and the number of unpaired electrons. (Atomic number of Fe = 26) (*Analyse*) [2]

(c) Give one use of EDTA in medicine. (*Recall*)

[1]

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